C-HI Configuration Lock

Patrick/Stas feedback applied before Kunanyi gate

Barry W. Brook and Codex

2026-05-08

Table of contents

Purpose 1

Decisions Applied 1

Grid 2

Output And Restart 2

Static Validation 3

Next Gate 3

Purpose

This note records the revised `C-HI` configuration after Patrick and Stas reviewed the first supervisor packet. It is now the working configuration for the next `kunanyi` gate, subject to live scheduler/build validation.

The tracked artifacts are under:

```
docs/science/assets/c_hi_configuration_lock_2026-05-08/
```

Decisions Applied

Topic	Locked value
Case	`C-HI`, Stewart cluster `002_003`
Environment	radially averaged tabulated file `tabulated_cosmo_002003.csv`
Physics	static-grid HD, 3D Cartesian
Jet power	`1e45 erg/s`, one-sided
Jet speed	`0.1 c`

Topic	Locked value
Opening angle	`15 deg` primary and secondary
`JET_RELATIVISTIC_DENSITY`	`NO`
`JET_INITIAL_RADIUS`	`0.1 kpc`
`JET_INJECTION_HEIGHT`	`0.75 kpc`
`JET_CHI`	`1.0` pressure-matched
First run target	`50 Myr`
First view	`0 deg`

Grid

Stas requested extending the jet axis to `X3 = +/-250 kpc` while preserving the resolution at `200 kpc` across coordinates. The rendered grid is:

```
X1-grid     5   -200 330 s -10 100 s -2.5 100 u 2.5 100 s 10 330 s 200
X2-grid     5   -200 330 s -10 100 s -2.5 100 u 2.5 100 s 10 330 s 200
X3-grid     5   -250 357 s -10 100 s -2.5 100 u 2.5 100 s 10 357 s 250
```

The old `10-200 kpc` outer stretch patch has `dx ~ 1.6923 kpc` at `200 kpc`. Using `357` outer cells in the `10-250 kpc` X3 patch gives `dx ~ 1.6930 kpc` in the cell spanning `200 kpc`.

Output And Restart

Patrick clarified that this static-grid PLUTO build restarts from `*.dbl` or `*.dbl.h5`, not Chombo checkpoint files. The rendered output plan is therefore:

Output	Cadence	Role
`flt.h5`	`1 Myr` (`306.60139379`)	science snapshots
`dbl.h5`	`5 Myr` (`1533.00696894`)	restart-capable double output
`particles_dbl`	`0.01 Myr` (`3.06601394`)	high-time-resolution particle proxy
`Checkpoint_interval`	disabled (`-1.0`)	not used for static-grid PLUTO

The particle-enabled run injects `16` particles every `0.01 Myr`:

```
Nparticles          -1    16.0
PART_INJECT_FREQUENCY       0.01
```

The `kunanyi` PBS template now passes PLUTO a walltime-safe maximum runtime:

```
./pluto -i pluto.ini -e environment.ini -maxtime 47.5
```

For restarts from HDF5 double outputs, the template detects `dbl.h5.out` and uses:

```
./pluto -h5restart <n>
```

Static Validation

`config/render_validation.json` reports `ok = true`.

Check	Expected	Rendered
`tstop`	15330.06968940	15330.06968940
`flt.h5`	306.60139379	306.60139379
`dbl.h5`	1533.00696894	1533.00696894
`Checkpoint_interval`	-1.0	-1.0
`particles_dbl`	3.06601394	3.06601394
`Nparticles` second field	16.0	16.0
`PART_INJECT_FREQUENCY`	0.01	0.01
`JET_CHI`	1.0	1.0
`JET_INITIAL_RADIUS`	0.1	0.1
`JET_INJECTION_HEIGHT`	0.75	0.75

Next Gate

The next live step is a `kunanyi` no-jet Stewart tabulated environment/MPI/HDF5 gate using the updated assets. After that is green, the first jet-bearing run should be an early, restartable, particles-enabled `C-HI` segment. The first inspection point remains the first healthy `3-5` snapshots, with density, pressure, tracer, and jet-axis velocity diagnostics before continuing.